

# Origins of Matrix/Vector and Matrix/Matrix Multiplication

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## 1 Generalizing $a x = b$

Why is matrix/vector and matrix/matrix multiplication defined the way it is? One motivation is to try to extend the (albeit trivial) solution of linear equations in 1-dimension to  $n$  dimensions. To do this, we go through the solution of the 1-dimensional case in careful detail.

The scalar problem is:

$$a x = b \tag{1}$$

Here is a very explicit solution keeping an eye towards generalization.

$$a x = b \tag{2}$$

$$a^{-1}(a x) = a^{-1}b \quad (\text{Multiply by Inverse}) \tag{3}$$

$$(a^{-1}a)x = a^{-1}b \quad (\text{Use Associativity of multiplication}) \tag{4}$$

$$1 x = a^{-1}b \quad (\text{What Inverse multiplication does}) \tag{5}$$

$$x = a^{-1}b \quad (\text{What Identity multiplication does}) \tag{6}$$

Keep this five-line ritual in view; it is the engine of the whole paper. The strategic move is one the reader has used since grade school: *isolate and invert* – get the unknown under a single operation, then apply that operation’s inverse. Our plan is to play the *same game on a bigger board*: replace the numbers by vectors and rectangular arrays of coefficients, and then ask what “multiply,” “associate,” “identity,” and “inverse” must *mean* for the five lines to remain legal. The payoff of this approach is that the definitions of matrix/vector and matrix/matrix multiplication will not be conventions handed down to be memorized; each will be *forced*.

The multi-dimensional problem has a lot more variables and coefficients. To start, we need to be more systematic about the naming of these coefficients. With this in mind, the multi-dimensional linear problem

can be written:

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2 \\ &\vdots \\ a_{n1}x_1 + a_{n2}x_2 + \cdots + a_{nn}x_n &= b_n \end{aligned}$$

Here,  $a_{ij}$  is the coefficient in the  $i^{\text{th}}$  row and  $j^{\text{th}}$  column.

To make this look like the 1-dimensional case, we need to think of the  $b$ 's as a single unit. Our single unit will be the *vector* of the  $b$ 's.

The multi-dimensional case can be rewritten in vector terms as:

$$\begin{bmatrix} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n \\ \vdots \\ a_{n1}x_1 + a_{n2}x_2 + \cdots + a_{nn}x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix} \quad (7)$$

We can replace the tedious list of sums on the left hand side of the above with standard mathematical summing notation:

$$\begin{bmatrix} \sum_{j=1}^n a_{1j}x_j \\ \sum_{j=1}^n a_{2j}x_j \\ \vdots \\ \sum_{j=1}^n a_{nj}x_j \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix} \quad (8)$$

Two things need to be done: treat the  $x$ 's as a unit – as we did with the  $b$ 's – and separate the  $a$  coefficients from the  $x$ 's. This must be done formally while preserving the meaning of the original problem. This is done in the most natural of ways:

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix} \quad (9)$$

Taking the rectangular collection (matrix) of  $a$ 's to be represented by  $A$ ; the (vector) collection of  $x$ 's to be  $\mathbf{x}$ ; and the collection of the  $b$ 's to be  $\mathbf{b}$ ; we may write (9) as:<sup>1</sup>

$$A \mathbf{x} = \mathbf{b} \quad (10)$$

Here,  $A$  represents the “matrix” of the  $a$ 's, with  $A_{i,j} = a_{ij}$ ; meaning, that the  $i^{\text{th}}$  row and  $j^{\text{th}}$  column of  $A$  is  $a_{ij}$ . This now looks like the scalar problem, (1). Just as one writes  $ax = b$ , interpreting the proximity of  $a$  and  $x$  to mean multiplication, we want the proximity of  $A$  and  $\mathbf{x}$  to mean matrix/vector multiplication – something we need to work out.

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<sup>1</sup>In general, we use uppercase italic letters for matrices; lowercase bold roman letters for vectors; and lowercase italic letters for scalars.

## 2 Solution of $A\mathbf{x} = \mathbf{b}$ determines Matrix Multiplication

We proceed to try solving (10) using the solution procedure used above for the scalar case. In the process, we need to:

- Vectorize the input variables,  $x_i$ , and the outputs,  $b_i$ . – **Done**.
- Define an “ $a$ ” matrix. – **Done**.
- Define matrix-vector multiplication.
- Define matrix-matrix multiplication.
- Define the identity matrix.
- Define the inverse matrix.

Here, in our new notation, is what the solution outline looks like for the multi-dimensional problem:

$$A\mathbf{x} = \mathbf{b} \tag{11}$$

$$A^{-1}(A\mathbf{x}) = A^{-1}\mathbf{b} \quad (\text{Matrix/vector multiply the vector equation by } A \text{ Inverse}) \tag{12}$$

$$(A^{-1}A)\mathbf{x} = A^{-1}\mathbf{b} \quad (\text{Use Associativity of matrix/matrix multiplication}) \tag{13}$$

$$I\mathbf{x} = A^{-1}\mathbf{b} \quad (\text{What Inverse matrix/matrix multiplication does}) \tag{14}$$

$$\mathbf{x} = A^{-1}\mathbf{b} \quad (\text{What Identity matrix/vector multiplication does}) \tag{15}$$

How do we make sense of this matrix/vector syntax on the left hand side of (11)? It should have the proper meaning; that is, (9) should have the same meaning as (8).

Examining (9) and the left hand side of (8), gives us our definition of matrix/vector multiplication:

$$[A\mathbf{x}]_i \equiv \sum_{j=1}^n A_{ij}x_j \tag{16}$$

That is,  $A$  acts on a vector,  $\mathbf{x}$ , to create a new vector,  $A\mathbf{x}$ , whose  $i^{\text{th}}$  entry is the  $i^{\text{th}}$  entry of the left hand side of (8).

Another way to write this is:

$$A\mathbf{x} = \sum_{j=1}^n x_j \mathbf{A}^j \tag{17}$$

Here,  $\mathbf{A}^j$  is the  $j^{\text{th}}$  column vector of  $A$ .

This can be seen by writing (7) as:<sup>2</sup>

$$x_1 \begin{bmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{n1} \end{bmatrix} + x_2 \begin{bmatrix} a_{12} \\ a_{22} \\ \vdots \\ a_{n2} \end{bmatrix} + \cdots + x_n \begin{bmatrix} a_{1n} \\ a_{2n} \\ \vdots \\ a_{nn} \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix} \quad (18)$$

From this we can see that matrix/vector multiplication of  $A$  on  $\mathbf{x}$  is a “weighted” combination of the columns of  $A$  using the elements of  $\mathbf{x}$  as the weights. Notice that if we pick an “ $\mathbf{x}$ ” with all of the weight concentrated on the  $j^{\text{th}}$  entry, then matrix/vector multiplication on this specific vector yields the  $j^{\text{th}}$  column of  $A$ ,  $\mathbf{A}^j$ . Specifically, applying  $A$  to the special vectors,  $\mathbf{e}_j$  – which are 0 everywhere except at  $j$  where they are 1 – we see that  $A \mathbf{e}_j = \mathbf{A}^j$ . Consequently, matrix/vector multiplication on this set of vectors uniquely determines  $A$  – if there is another matrix that gives the same result for these vectors with respect to matrix/vector multiplication, its columns would have to match  $A$ .

We note the following for future reference.

$$[\mathbf{A}^j]_i = A_{i,j} \quad (19)$$

That is, the  $i^{\text{th}}$  entry of the column vector,  $\mathbf{A}^j$ , is the  $i^{\text{th}}$  row,  $j^{\text{th}}$  column of the matrix  $A$ .

Before moving on with our outline, we note one very important property of matrix/vector multiplication: It is *linear*. By that we mean the following:<sup>3</sup>

$$A(\alpha \mathbf{x} + \beta \mathbf{y}) = \alpha A\mathbf{x} + \beta A\mathbf{y} \quad (20)$$

One can show this from our definition of matrix/vector multiplication which we leave to the reader. What this means in practice is that for a given vector,  $\mathbf{z} = \alpha \mathbf{x} + \beta \mathbf{y}$ , we can compute  $A\mathbf{z}$  by computing  $A$  on the “components” of  $\mathbf{z}$  and then multiplying by scalars and adding the resulting vectors. This can be an easier way to compute the *action* of a matrix on a vector in some cases. The linearity result above can be extended to any finite linear combination:

$$A \left( \sum_{j=1}^m c_j \mathbf{x}_j \right) = \sum_{j=1}^m c_j A\mathbf{x}_j \quad (21)$$

Between them, the last two observations – the  $\mathbf{e}_j$  pick off columns, and linearity extends a fact about the  $\mathbf{e}_j$  to a fact about every vector – form a move we will replay several times below. Call it *probe with the standard basis, then extend by linearity*. We have just used the probing half to show that a matrix is completely determined by its action on the  $\mathbf{e}_j$ ; the same probe will shortly force the formula for matrix/matrix multiplication and pin down the identity matrix, and the extend-by-linearity half will each time upgrade what happens on the  $\mathbf{e}_j$  to what happens everywhere.

<sup>2</sup>Here, we assume the reader has a passing knowledge of how vectors are added and multiplied by scalars.

<sup>3</sup>You might ask: “How did someone think up such a strange property?” But it is not strange, it is what one might hope for, as all we are saying is that matrix/vector multiplication works the same way that ordinary *scalar* multiplication does – it *distributes* through addition and scalar multiplication.

Continuing with our solution outline, matrix/matrix multiplication is needed to go from equation (12) to (13). How must this be defined? Well, we need to make sense of:

$$(A^{-1}A)\mathbf{x} = A^{-1}(A\mathbf{x}) \quad (22)$$

This involves the inverse of the matrix  $A$ , which we also need to define. We want associativity to hold *for all* matrix products, not just  $A^{-1}A$ . So we forgo what the inverse of a matrix might mean and focus on imposing the condition that matrix/matrix multiplication be associative for *all* input vectors. Specifically, the condition we impose is:

$$(BA)\mathbf{x} = B(A\mathbf{x}) \quad \forall \mathbf{x} \in \mathbf{R}^n \quad (23)$$

Notice that we are imposing what matrix/matrix multiplication is by saying how the new matrix formed,  $BA$ , acts on an *arbitrary* vector,  $\mathbf{x}$ . And we specify that action by the right hand side of (23), which involves only matrix/vector multiplication – something we have defined.

We want to emphasize that this is a strong condition to impose; meaning, that the requirement that this is true for *all*  $\mathbf{x}$  gives us a lot to work with. For instance, if one had two  $n \times n$  matrices,  $C$  and  $D$ , what could you conclude if someone told you that  $C\mathbf{x} = D\mathbf{x}$  for *some*  $n$  vector,  $\mathbf{x}$ ? Well, the answer is: not much. However, if I told you that  $C\mathbf{x} = D\mathbf{x}$  *for all*  $\mathbf{x}$ , what could you say? I claim that tells us that  $C \equiv D$ ; meaning that every entry in  $C$  is the same as the corresponding entry in  $D$ . We can see this by applying  $C$  and  $D$  to each of the  $\mathbf{e}_j$ , ( $j \in [1, n]$ ) vectors. For each  $j$ ,  $C\mathbf{e}_j$  and  $D\mathbf{e}_j$  pick off the  $j^{\text{th}}$  column of  $C$  and  $D$  respectively. So we see that each column of  $C$  and  $D$  must match; consequently,  $C \equiv D$ .

We apply this idea to (23) to find a formula for matrix/matrix multiplication. Since (23) must be true *for all*  $\mathbf{x}$ ; in particular, it must be true for the vectors  $\{\mathbf{e}_j\}_{j=1}^n$  – the probing half of our move. This gives us the following equations:

$$(BA)\mathbf{e}_j = B(A\mathbf{e}_j) \quad j \in [1, n] \quad (24)$$

Or,

$$(BA)^j = B\mathbf{A}^j \quad j \in [1, n] \quad (25)$$

For any given,  $i$ , the  $i^{\text{th}}$  entry of the left and right hand side is (after using the definition of matrix/vector multiplication, (16), on the right hand side)

$$[(BA)^j]_i = \sum_{k=1}^n B_{i,k} [\mathbf{A}^j]_k \quad \forall i, j \in [1, n] \quad (26)$$

Using (19) on the left and right hand sides of the previous equation we have

$$(BA)_{i,j} = \sum_{k=1}^n B_{i,k} A_{k,j} \quad \forall i, j \in [1, n] \quad (27)$$

And we are done: we have shown what the new matrix  $BA$  is by giving every entry  $(BA)_{i,j}$ . So using the set of vectors,  $\{\mathbf{e}_j\}_{j=1}^n$ , we have completely determined what the matrix multiplication of  $BA$  is in order that matrix/matrix multiplication be associative when using these vectors. Does this mean that associativity

works for *arbitrary* vectors? The answer is yes, by the linearity of matrix/vector multiplication – the extend-by-linearity half of our move. Any given  $\mathbf{x}$  can be written as a *linear combination*<sup>4</sup> of the vectors  $\{\mathbf{e}_j\}_{j=1}^n$ :  $\mathbf{x} = a_1\mathbf{e}_1 + a_2\mathbf{e}_2 + \cdots + a_n\mathbf{e}_n$ . Matrix multiply the left and right hand sides by the matrix,  $BA$ . This gives

$$\begin{aligned}
(BA)\mathbf{x} &= (BA) \left( \sum_{j=1}^n a_j \mathbf{e}_j \right) && \text{Replace } \mathbf{x} \text{ with its components, } \mathbf{e}_j. \\
&= \sum_{j=1}^n a_j (BA)\mathbf{e}_j && \text{Linearity of matrix/vector multiplication by } (BA) \text{ (itself a matrix).} \\
&= \sum_{j=1}^n a_j B(A\mathbf{e}_j) && \text{Associativity of matrix/matrix multiplication on components, } \mathbf{e}_j. \\
&= B \left( \sum_{j=1}^n a_j A\mathbf{e}_j \right) && \text{Linearity of matrix/vector multiplication by } B. \\
&= B \left( A \left( \sum_{j=1}^n a_j \mathbf{e}_j \right) \right) && \text{Linearity of matrix/vector multiplication by } A. \\
&= B(A\mathbf{x}) && \text{Replace components with } \mathbf{x}.
\end{aligned} \tag{28}$$

In the above calculations, we applied our new matrix,  $(BA)$  to an arbitrary  $\mathbf{x}$ , and used the *linearity* of matrix/vector multiplication and the *associativity* of matrix/matrix multiplication on the components,  $\{\mathbf{e}_j\}_{j=1}^n$ , to show that matrix/matrix multiplication, as we have defined it, is associative – *irrespective* of the input vector.

The next order of business is to identify an  $n \times n$  matrix which serves as an identity (in terms of matrix/vector multiplication). Let us suppose that we have such a matrix and let's call it  $I$ . Then we must have  $I\mathbf{x} = \mathbf{x} \quad \forall \mathbf{x} \in \mathbf{R}^n$ . Then, in particular (probing with the standard basis once more),  $I\mathbf{e}_j = \mathbf{e}_j \quad \forall j \in [1, n]$ . However, we know that  $I\mathbf{e}_j = \mathbf{I}^j$ . Therefore,  $I$  must have the property that  $\mathbf{I}^j = \mathbf{e}_j$ . Consequently,  $I$  must have the form:

$$I = \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \\ 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ & & & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \end{bmatrix} \tag{29}$$

We have shown that the only candidate for the identity matrix with respect to matrix/vector multiplication is the matrix  $I$ . That is, if there is an identity matrix, it must be the matrix of equation (29). Does it satisfy the property of being an identity matrix (again, in the matrix/vector multiplication world)? That is, do we have  $I\mathbf{x} = \mathbf{x}$  for all  $\mathbf{x} \in \mathbf{R}^n$ ? As before, we can express  $\mathbf{x}$  as:  $\mathbf{x} = \sum_{j=1}^n c_j \mathbf{e}_j$ . By the *linearity* of

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<sup>4</sup>Again, we assume the reader is familiar with expanding a vector in terms of the standard basis vectors.

matrix/vector multiplication, we have

$$\begin{aligned}
I\mathbf{x} &= I \left( \sum_{j=1}^n c_j \mathbf{e}_j \right) && \text{Replace } \mathbf{x} \text{ with components, } \mathbf{e}_j. \\
&= \sum_{j=1}^n c_j I\mathbf{e}_j && \text{Linearity of matrix/vector multiplication.} \\
&= \sum_{j=1}^n c_j \mathbf{e}_j && \text{Identity of } I \text{ on components, } \mathbf{e}_j. \\
&= \mathbf{x} && \text{Replace components with } \mathbf{x}.
\end{aligned} \tag{30}$$

One can show that this identity matrix,  $I$ , is also the identity operator for matrix/matrix multiplication: applying  $(IA)\mathbf{e}_j = I(A\mathbf{e}_j) = A\mathbf{e}_j$  to each  $j$  shows  $(IA)^j = A^j$  for all  $j$ , hence  $IA = A$ ; the same argument on the columns of  $A$  gives  $AI = A$ .

The only remaining piece of the ritual is the inverse: when does a matrix  $A^{-1}$  exist, and how might one compute it? We do not attempt this here – it is genuine unfinished business, and we flag exactly where the difficulty lives: an inverse must undo  $A$  on *every* vector, and whether that is possible depends on whether the columns of  $A$  are rich enough to reach all of  $\mathbf{R}^n$  (a first taste of this appears in the next section). The companion paper “What is a Determinant” builds the tool that detects the failure: a single number, the signed factor by which  $A$  scales volume, whose multiplicative property shows that if  $A^{-1}$  exists then  $|A||A^{-1}| = |I| = 1$  – so a matrix with determinant 0, one that flattens the unit cube, can have no inverse.

Here is the state of the dictionary between the scalar world and its vectorized cousin:

| Scalar world                           | Matrix world                                         | Status                                                   |
|----------------------------------------|------------------------------------------------------|----------------------------------------------------------|
| $ax = b$                               | $A\mathbf{x} = \mathbf{b}$                           | the problem, repackaged                                  |
| product $ax$                           | $[A\mathbf{x}]_i = \sum_j A_{ij}x_j$                 | forced: must preserve the equations                      |
| $(ba)x = b(ax)$                        | $(BA)\mathbf{x} = B(A\mathbf{x})$ , all $\mathbf{x}$ | imposed; forces $(BA)_{ij} = \sum_k B_{ik}A_{kj}$        |
| the number 1                           | the matrix $I$                                       | forced: $I\mathbf{e}_j = \mathbf{e}_j$ picks its columns |
| $a^{-1}$ (exists iff $a \neq 0$ )      | $A^{-1}$                                             | deferred: see “What is a Determinant”                    |
| range of $a$ : $\{0\}$ or $\mathbf{R}$ | range of $A$ : span of its columns                   | next section                                             |

With the dictionary in hand, we continue with a qualitative comparison of the solution to the scalar problem,  $ax = b$ , and its vectorized cousin.

### 3 Qualitative Features of Solutions

We can view the multiplication of two numbers,  $a$  and  $x$ , as just that. Or, we can think of  $a$  being fixed and letting  $x$  “run through” all numbers. Here we see two cases: if  $a \neq 0$ , then letting  $x$  run through all of the numbers in  $\mathbf{R}$  will produce all of the numbers in  $\mathbf{R}$ . We could think of  $a$  as an “operator” and call the set of all possible outputs, the range of  $a$  and denote it:  $\mathcal{R}(a)$ . There is another case:  $a$  could be zero. In this

case its range is the set  $\{0\}$ . In the first case, with non-zero  $a$  it is clear that we can find an  $x$  to “hit” a given value  $b$ . That is, we can solve  $ax = b$ .

One can define the same concept for a matrix,  $A$ . In this case, we produce a “space” of all possible outputs of  $A$  as it acts on every input vector  $\mathbf{x} \in \mathbf{R}^n$ . This amounts to, using (17), the set of all *linear combinations* of the columns of  $A$ . Using this language of ranges, here is what we can say about the scalar problem:  $ax = b$ .

**Unique Solution:** If  $a^{-1}$  exists ( $a \neq 0$ ), then there is a **unique** solution.

**No Solution:** If  $a^{-1}$  does not exist (i.e.,  $a = 0$ ) *AND*  $b$  is **not** in the range of  $a$  (that is:  $b \neq 0$ ), then there is **no** solution.

**Infinite Solutions:** If  $a^{-1}$  does not exist (i.e.,  $a = 0$ ) *BUT*  $b$  is in the range of  $a$  (that is:  $b = 0$ ), then there are an **infinite** number of solutions.

Here is the analog of this solution categorization for the multi-dimensional case:  $A\mathbf{x} = \mathbf{b}$ .

**Unique Solution:** If  $A^{-1}$  exists, then there is a **unique** solution.

**No Solution:** If  $A^{-1}$  does not exist *AND*  $\mathbf{b}$  is not in the range of  $A$  (that is  $\mathbf{b} \notin \mathcal{R}(A)$ ), then there is **no** solution.

**Infinite Solutions:** If  $A^{-1}$  does not exist *BUT*  $\mathbf{b}$  is in the range of  $A$  (that is:  $\mathbf{b} \in \mathcal{R}(A)$ ), then there are an **infinite** number of solutions.

## 4 Curtain Call

We can now answer the opening question in the paper’s own words. Matrix/vector multiplication is defined the way it is because it is the only definition under which a rectangular array of coefficients can act on the unknowns-as-a-unit while *preserving the meaning* of the original system of equations. Matrix/matrix multiplication is defined the way it is because it is the only definition under which the five-line scalar ritual – multiply by the inverse, reassociate, cancel – remains legal on the bigger board: we *imposed* associativity for all inputs, probed with the standard basis, and the row-by-column recipe  $(BA)_{ij} = \sum_k B_{ik}A_{kj}$  fell out. The recipe usually handed down as a convention is not one; it is what associativity looks like in coordinates. And the game itself – isolate and invert – is unchanged from the one-dimensional case; only the board grew.